

Total No. of Questions : 8]

SEAT No. :

PA-1623

[Total No. of Pages : 2

[5926]-257

T.E. (Computer Engineering) (Honors/Minors)

DATA SCIENCE

DATA SCIENCE AND VISUALIZATION

(2019 Pattern) (Semester - I) (310501)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) What is Linear regression? List the applications where linear regression can be applied? [6]

b) State and explain how Naïve Bays classifier can be used to solve the classification problems? [6]

c) Write a note on association rules. [6]

OR

Q2) a) What is clustering? Explain K-means clustering algorithm. [6]

b) Explain Apriori Algorithm used in machine learning with valid example. [6]

c) Illustrate how will you evaluate association rules. [6]

Q3) a) State and explain the different constituents of the decision tree. [9]

b) Write a note on the perceptron model. [8]

OR

Q4) a) When do you use Backpropagation in Neural Networks? Explain by taking a suitable example. [9]

b) What is entropy? How entropy is calculated explain with a suitable example. [8]

P.T.O.

- Q5)** a) Define the term Dashboard along with its evolution and steps to design the dashboard. [9]
- b) Write a note on: [9]
- i) Pie charts
 - ii) Bar graphs
 - iii) Scatterplots

OR

- Q6)** a) Explain the terms Network hierarchies and reports associated with data visualization. [6]
- b) Write a note on advanced visualization techniques and explain anyone of them. [6]
- c) Write a note on 'display media for Dashboard. [6]
- Q7)** a) What are different types of data model explain in brief. [6]
- b) List the advantages of multi-dimensional data model? [6]
- c) Discuss the challenges of clustering High-dimensional data. [5]

OR

- Q8)** a) Explain the need of data modelling. [6]
- b) Explain multidimensional data model with one example. [6]
- c) What do you mean by Principal Component Analysis? [5]

